

Álvaro López Álvarez

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LinkedIn, GitHub, Portfolio

Backend Developer with a Master's Degree in Computer Engineering (8.8/10) focused on distributed systems. Through self-directed projects, has built Java microservices with Spring Boot and Apache Kafka, RESTful APIs with JSON exchange over PostgreSQL, and Docker-based environments, backed by unit and integration tests with JUnit and Testcontainers. Two roles at the Instituto Tecnológico de Informática (ITI): React/TypeScript features integrated with back-end services, and validated read-only SQL over a PostgreSQL data-governance schema to resolve data-access policies. Comfortable collaborating with front-end teams on API integration.

Core Competencies

- **Java & Spring**: Java, Spring Boot 3, JPA, Jakarta Validation, RESTful APIs, JSON, DTOs/mappers, centralized error handling
- **Microservices & Messaging**: Apache Kafka (outbox pattern, dead-letter topics, idempotent consumers), NATS JetStream, Apache APISIX, OAuth 2.0/JWT
- **Testing**: JUnit, Testcontainers, unit and integration tests, TDD
- **Databases & DevOps**: PostgreSQL, SQL, Firebase Realtime Database (NoSQL), Supabase, Docker, Docker Compose, Git, GitHub Actions
- **Other Languages**: Golang, TypeScript/JavaScript (React), Python, Kotlin

Professional Experience

Instituto Tecnológico de Informática (ITI) — **Valencia, España**
R&D Developer — *Feb 2026–Jun 2026*

- Built a LangGraph multi-step agent that converts natural-language questions into validated, read-only SQL queries over a PostgreSQL data-governance schema to resolve data-access policies
- Designed the workflow for safety and robustness: schema/table selection, structured SQL generation, execution validation, and retry-on-error
- Implemented LLM-based improvements to a search engine within an application for the management, governance, and sovereignty of industrial data

Instituto Tecnológico de Informática (ITI) — **Valencia, España**
Full-Stack Developer Intern — *Feb 2024–Jun 2024*

- Developed and maintained the corporate personnel management web application (Intranet)
- Implemented new front-end features using React and TypeScript, improving internal usability
- Contributed to back-end development, ensuring smooth integration between services

Independent Java & Backend Projects

- **Kafka + Spring Boot Ticketing Platform** – *Java, Spring Boot 3, Apache Kafka, Testcontainers*: Event-driven microservices (reservation, payment, notification, audit) using the outbox pattern, dead-letter topics, and idempotent processing, with integration tests using Testcontainers.
- **English Learning API** – *Java, Spring Boot, JPA, Jakarta Validation*: RESTful API for practicing English phrasal verbs and collocations; JPA entities, DTOs/mappers, Jakarta validation, centralized error handling, and randomized practice challenges with attempt tracking.
- **HTTP from TCP (Java)** – *Java, JUnit*: HTTP/1.1 server built directly on raw sockets, covering Java I/O, concurrency with executors, and JUnit testing.
- **FaaS with Golang** – *Go, Docker, NATS JetStream, Apache APISIX*: Distributed Function-as-a-Service platform where workers execute arbitrary Docker-image functions. NATS JetStream work-queues for reliable task distribution, Apache APISIX gateway for JWT auth and rate limiting, horizontal scaling via Docker Compose replicas.

- **AIDucate** – *Go, Gin, PostgreSQL, React, TypeScript, Gemini API*: Full-stack SaaS platform generating personalized study plans, content, and exercises. JWT-authenticated REST API with PostgreSQL persistence; React/TypeScript frontend deployed on Vercel, backend on Render. *(In progress)*

Education

- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Master's Degree in Computer Engineering (MUIINF) *2024–2026*
  Completed (Sep 2026), final grade 8.8/10. Topics: distributed systems, high-performance computing, cybersecurity, embedded systems.
- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Grado (BSc) in Ingeniería Informática, Software Engineering track *2020–2024*
  GPA: 8.88/10. Thesis: "ValenMove – Unified Public Transport App" (grade 9/10), a native Android app (Kotlin, MVVM) integrating public transport APIs, web scraping, and Open Data for real-time arrivals.

Languages

- Spanish (native), Catalan (native), English (certified B2, confident conversational).

References

- Available upon request.